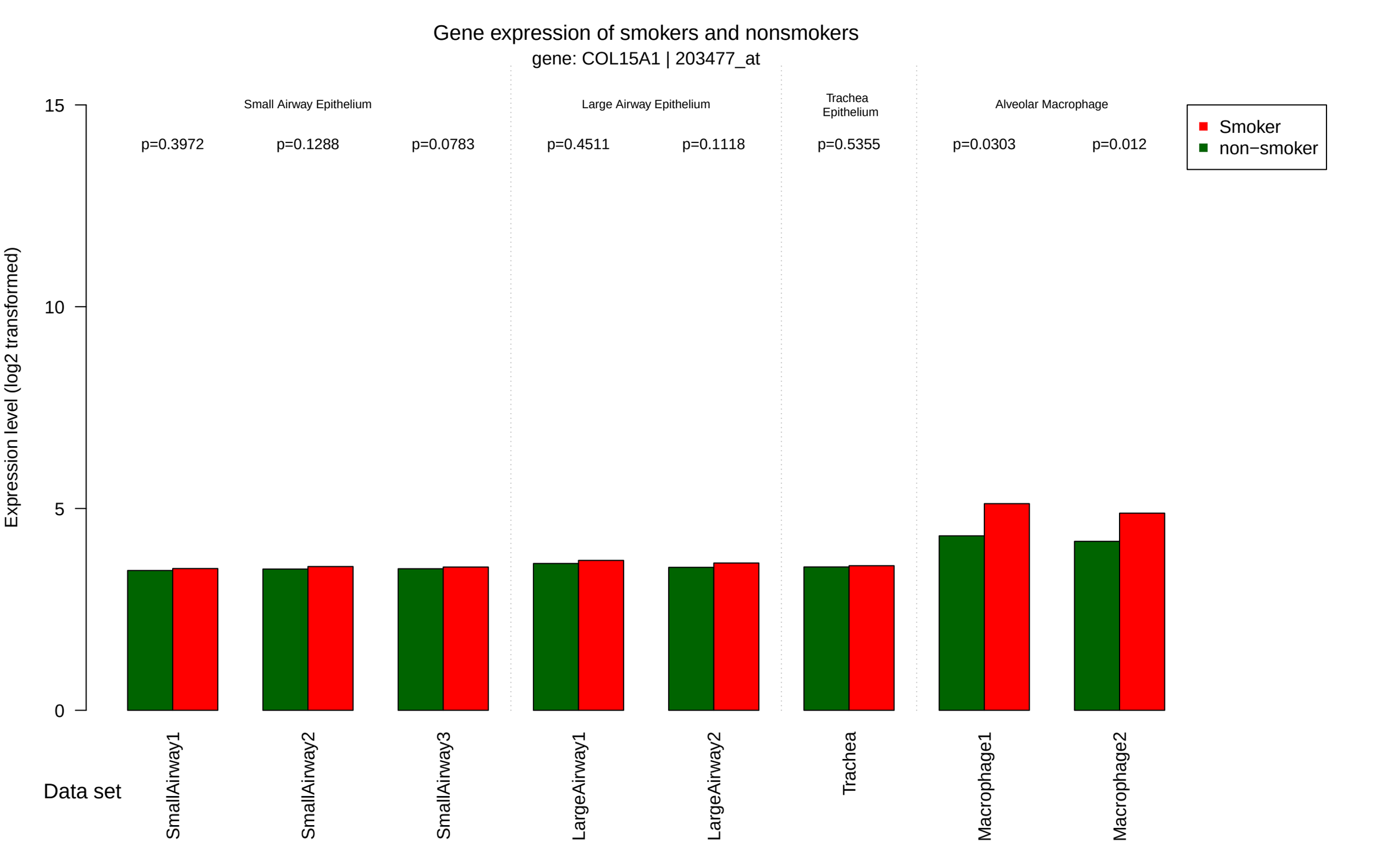
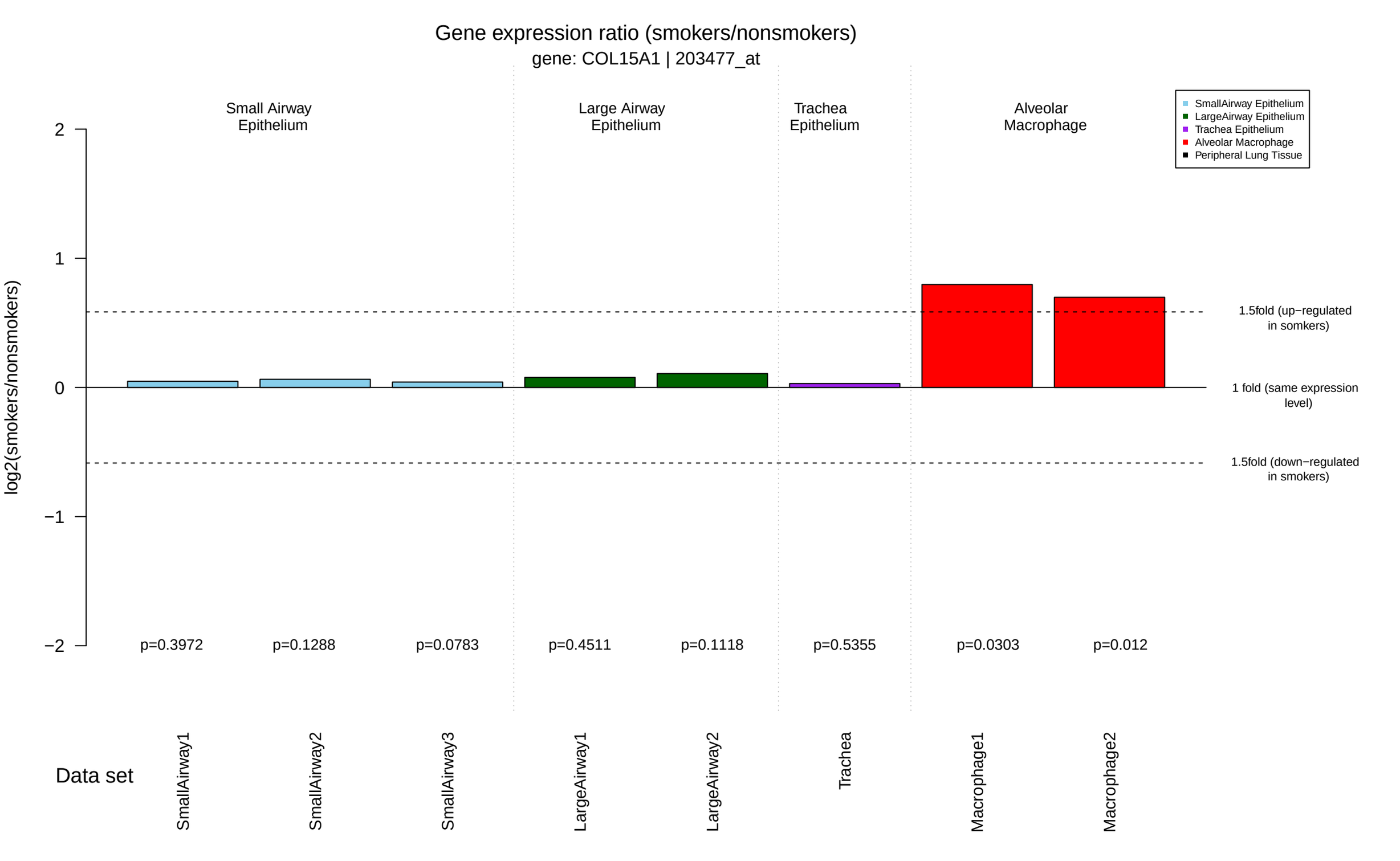
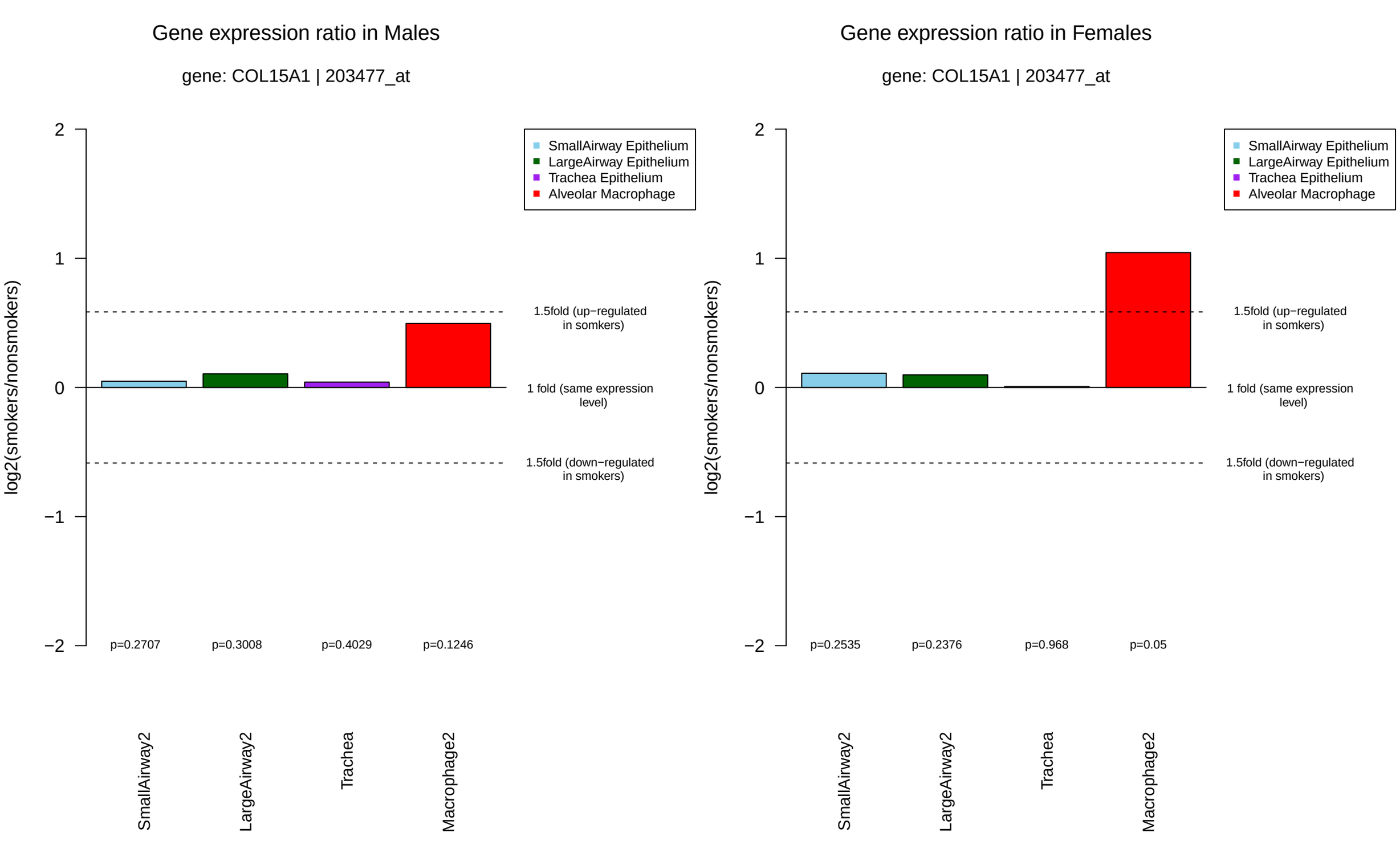
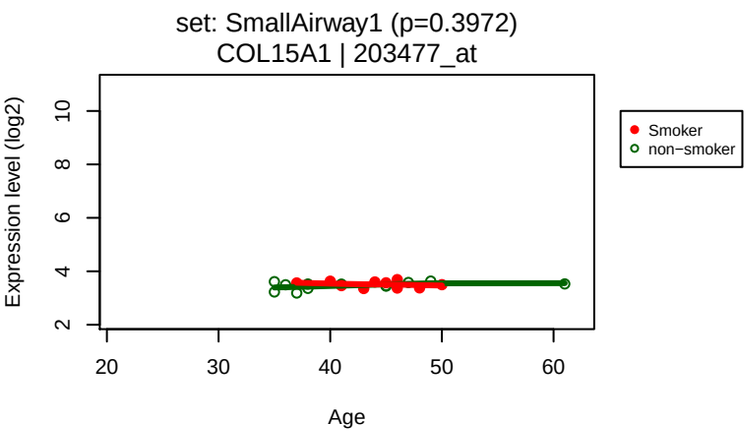
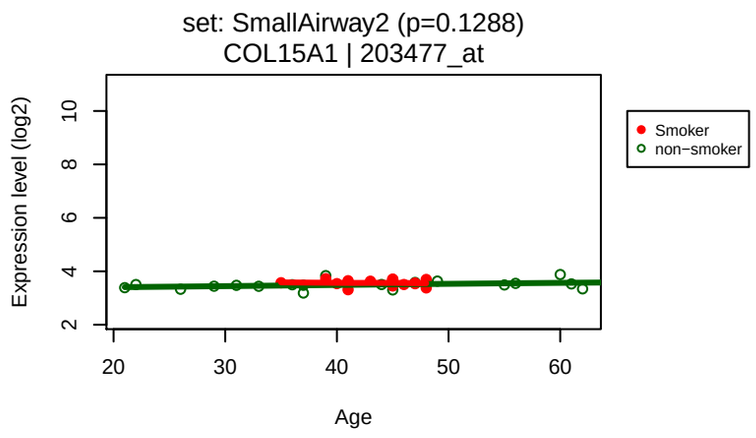
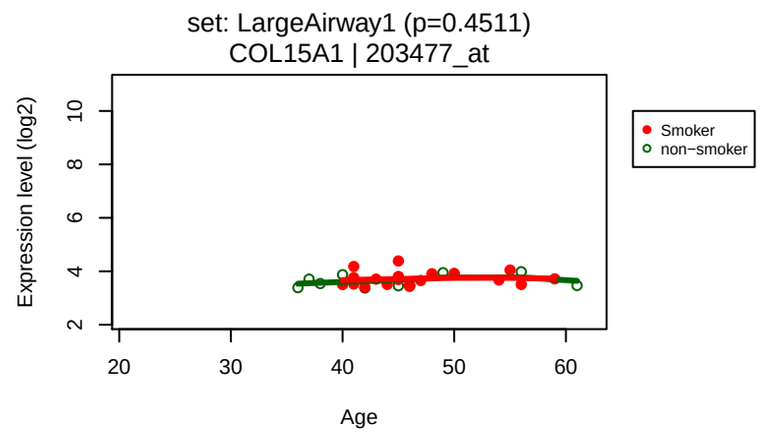

Gene expression of smokers and nonsmokers
gene: COL15A1 | 203477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3972
p=0.1288
p=0.0783
p=0.4511
p=0.1118
p=0.5355
p=0.0303
p=0.012
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

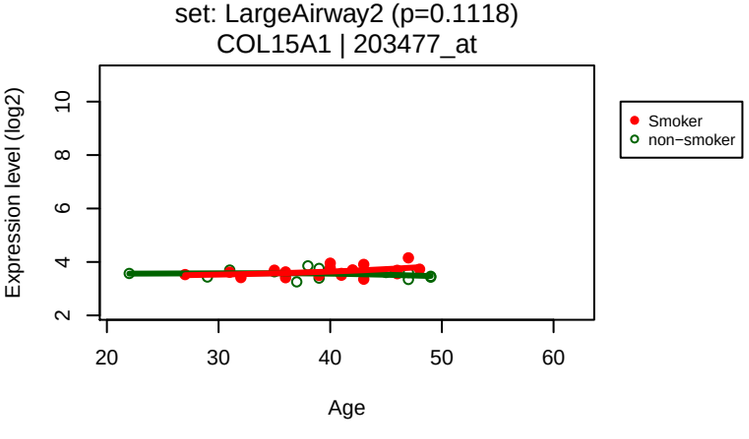

Gene expression ratio (smokers/nonsmokers)
gene: COL15A1 | 203477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3972
p=0.1288
p=0.0783
p=0.4511
p=0.1118
p=0.5355
p=0.0303
p=0.012
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

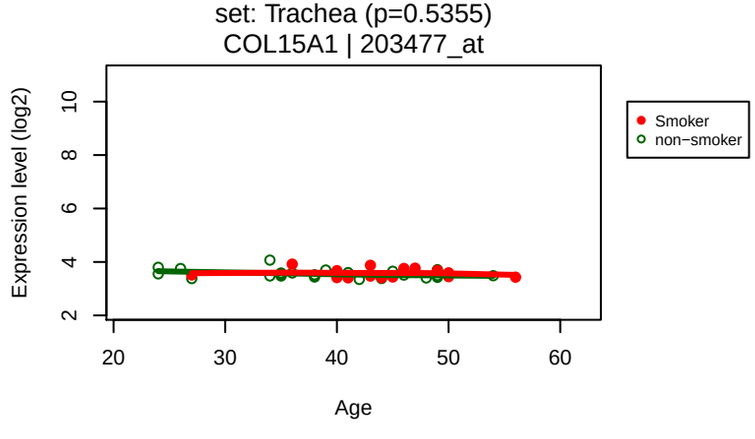

Gene expression ratio in Males
gene: COL15A1 | 203477_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2707
p=0.3008
p=0.4029
p=0.1246
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL15A1 | 203477_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2535
p=0.2376
p=0.968
p=0.05
SmallAirway2
LargeAirway2
Trachea
Macrophage2

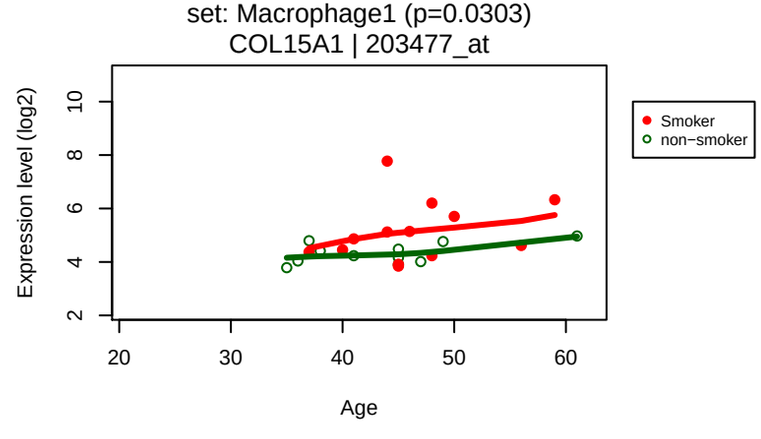

set: SmallAirway1 (p=0.3972)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker

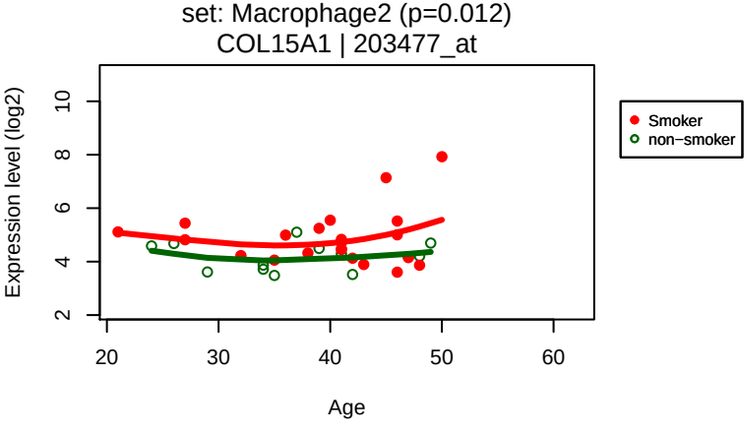

set: SmallAirway2 (p=0.1288)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker

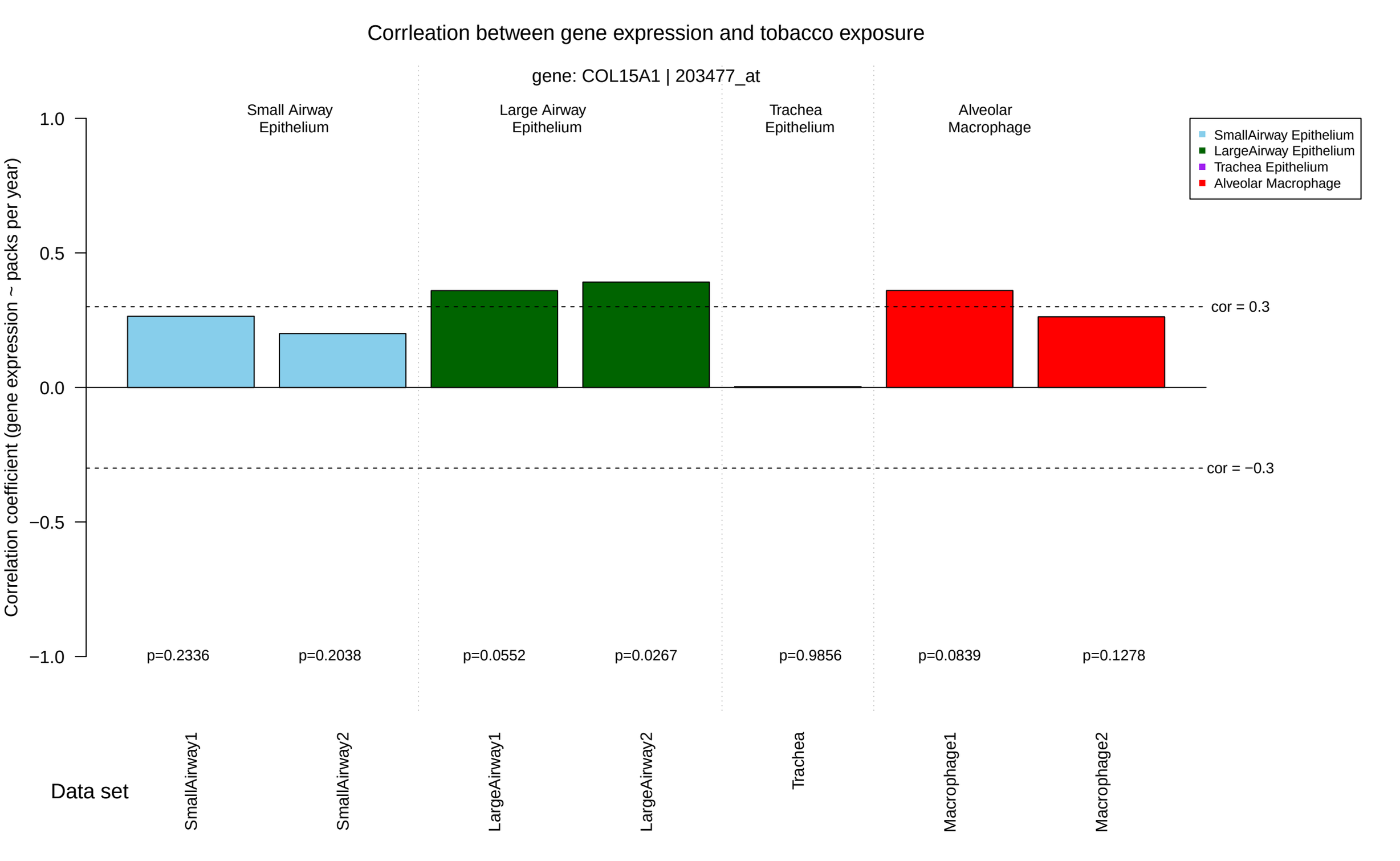

set: LargeAirway1 (p=0.4511)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1118)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5355)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0303)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.012)
COL15A1 | 203477_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL15A1 | 203477_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2336
p=0.2038
p=0.0552
p=0.0267
p=0.9856
p=0.0839
p=0.1278
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2